saying that there are a great deal of light packets – the electrons would not be extracted from their atoms if the individual grains of light are too small, that is, if the frequency of light is too low. This explains why it is the colour and not the intensity which determines whether the photoelectric effect occurs or not. For this simple reasoning Einstein was awarded the Nobel Prize. It is easy to understand things once someone has thought them through. The difficulty lies in thinking them through in the first place.

Today we call these packets of energy ‘photons’, from the Greek word for light: φώς. Photons are the grains of light, its ‘quanta’. In the article Einstein writes:

> It seems to me that the observations associated with blackbody radiation, fluorescence, the production of cathode rays by ultraviolet light, and other related phenomena connected with the emission or transformation of light are more readily understood if one assumes that the energy of light is discontinuously distributed in space. In accordance with the assumption to be considered here, the energy of a light ray spreading out from a point source is not continuously distributed over an increasing space but consists of a finite number of ‘energy quanta’ which are localized at points in space, which move without dividing, and which can only be produced and absorbed as complete units.[1]

These simple and clear lines are the real birth certificate of quantum theory. Note the wonderful initial ‘It seems to me …’, which recalls the hesitations of Faraday, or those of Newton; or the uncertainty of Darwin in the first pages of *On the Origin of Species*. True genius is aware of the momentousness of the steps it is taking, and is always hesitant …

There is a clear relation between Einstein’s work on Brownian motion (discussed in Chapter 1) and his work on the quanta of light, both completed in 1905. In the first, Einstein had managed to find a demonstration of the atomic hypothesis, that is to say, of the granular structure of matter. In the second he extends this same hypothesis to light: light must have a granular structure as well.

At first, Einstein’s idea that light could be made up of photons is regarded by his colleagues as no more than youthful waywardness. Everyone commends him for his theory of relativity, but everybody judges the notion of photons to be outlandish. Scientists had only recently been persuaded that light was a wave in the electromagnetic field: how could it be made up of grains? In a letter addressed to the German Ministry, recommending that Einstein should have a professorship inaugurated for him in Berlin, the most distinguished physicists of the day write that the young man is so brilliant that he ‘may be excused’ certain excesses, such as the idea of photons. Not many years later, the very same colleagues award him the Nobel Prize, precisely for having understood that photons exist. Light falls on a surface like a gentle hail shower.